

Deep Neural Networks

Course Project

Learning from Human Uncertainty

Predicting Annotator Disagreement on CIFAR-10

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Table of Contents

Introduction and Problem Statement

1.1 Key Contributions

Related Work

2.1 Crowdsourced Labels and Human Uncertainty

2.2 Loss Functions for Distribution Matching

2.3 Transfer Learning and Initialization

2.4 Evaluation Beyond Accuracy

2.5 Explainability: Grad-CAM

Dataset and Preprocessing

3.1 CIFAR-10 and CIFAR-10H

3.2 Dataset Architecture and Data Splits

3.3 Data Exploration and Human Disagreement Analysis

Model Architecture

4.1 Backbone - Modified ResNet-18

4.2 Backbone Initialization Strategy

4.3 Prediction Head

Training Protocol

5.1 Stage 1 - Hard-Label Pre-Training

5.2 Stage 2 - Soft-Label Fine-Tuning

Loss Functions

6.1 KL Divergence Loss (Baseline)

6.2 Jensen-Shannon Divergence Loss

6.3 Custom Entropy-Weighted Loss

7 Evaluation Metrics

Experimental Results

8.1 Stage 1 Training Dynamics

8.2 Loss Function Comparison - Fine-Tuning Convergence

8.3 Human Alignment Metrics - Quantitative Evaluation

8.4 Ablation Study - Backbone Initialization

9 Robustness Analysis

Explainability and Failure Analysis

10.1 Grad-CAM Analysis

10.2 Failure Case Analysis

10.3 Manual Disagreement Source Analysis

Discussion

11.1 Key Findings

11.2 Limitations

12 Conclusion

Appendix: Reproducibility

13.1 Dependencies and Environment

13.2 Execution Order

Abstract

Standard image classification pipelines discard information embedded in annotator disagreement by collapsing crowd-sourced responses into a single hard label. This project presents a two-stage deep neural network framework that explicitly models human uncertainty using CIFAR-10H, (a dataset of crowdsourced annotations for the 10,000-image CIFAR-10 test set). In Stage 1, a CIFAR-resolution-adapted ResNet-18 is pre-trained on the full 50,000-image CIFAR-10 training set using standard cross-entropy loss with hard labels. In Stage 2, the pre-trained backbone is fine-tuned on CIFAR-10H soft-label distributions under three competing loss functions: KL Divergence, Jensen-Shannon Divergence (JSD), and a novel Custom Entropy-Weighted Loss that focuses on ambiguous images during training. The backbone initialization ablation study demonstrates that pre-trained weights yield dramatically superior human uncertainty alignment. Evaluation is conducted through a five-dimensional metric suite covering distribution matching, entropy correlation, and retrieval quality (Precision@K). The central result is a Pearson entropy correlation of 0.84 with pre-trained initialization, compared to 0.42 with random initialization, confirming that prior representational learning is critical for effective uncertainty alignment.

Keywords: *soft labels, human uncertainty, CIFAR-10H, KL divergence, Jensen-Shannon divergence, entropy-weighted loss, Grad-CAM, failure analysis, Precision@K, ablation study*

1. Introduction and Problem Statement

Standard deep learning practice treats annotator disagreement as noise, resolving it through majority voting to produce a single hard class label. Our project deliberately inverts that assumption. The objective is to build a deep neural network that predicts the full probability distribution of human annotator responses for a given CIFAR-10 image, treating disagreement, *not* as an error to be discarded, rather as a meaningful and informative signal.

Where a conventional classifier might output "*cat*" or "*cat with 95% confidence*," our model is designed to output something richer, and more specific to distribution like :

[cat: 0.60, dog: 0.25, deer: 0.10, horse: 0.05, all others: 0.00]

This output encodes a prediction of how approximately 50 human annotators would collectively label the image, capturing both the dominant label and the pattern of confusion between classes. The central research question is therefore not "What is this image?" but "What will people think this image is, and how much will they disagree?"

This framing is motivated by real-world deployment contexts : medical imaging, content moderation, autonomous navigation, where communicating model uncertainty to human decision-makers is an engineering necessity, not an optional feature.

1.1 Key Contributions

- A correctly separated two-stage data architecture using the full 50,000-image CIFAR-10 training set for hard-label representational learning and the 10,000-image CIFAR-10H test set for soft-label fine-tuning.
- Three competing soft-label loss functions : KL Divergence, Jensen-Shannon Divergence, and a novel Custom Entropy-Weighted Loss, implemented as modular PyTorch nn.Module classes and compared head-to-head.
- A backbone initialization ablation study demonstrating that pre-trained weights double the Pearson entropy correlation (0.42 to 0.84) compared to random initialization.
- A five-dimensional evaluation suite covering distribution matching, entropy correlation, and Precision@K retrieval quality metrics.

- Systematic failure case analysis and Grad-CAM visualizations providing qualitative insight into residual uncertainty mismatches.

2. Related Work

2.1 Crowdsourced Labels and Human Uncertainty

Prior work on CIFAR-10H showed that training with soft labels yields better-calibrated predictions and model error patterns that more closely resemble human judgement. Human disagreement distributions have also been shown to predict which semantic categories are most difficult for neural networks. Separately, Northcutt et al. demonstrated that label noise and ambiguity are widespread in standard benchmark datasets, motivating probabilistic label representations over single hard-label annotations.

2.2 Loss Functions for Distribution Matching

KL divergence is the standard objective for matching predicted and target probability distributions and is widely used in soft-label learning settings such as knowledge distillation. However, its asymmetry and sensitivity to near-zero probabilities motivate alternative objectives such as Jensen–Shannon divergence, which provides a symmetric and bounded measure of distributional similarity. Our custom entropy-weighted loss further extends this framework by adaptively prioritizing examples with high annotator disagreement.

2.3 Transfer Learning and Initialization

Transfer learning has consistently been shown to improve downstream performance in data-limited regimes. In the context of disagreement prediction, pre-trained feature representations should enable the model to more effectively learn how visual ambiguity maps to human uncertainty. This hypothesis is evaluated directly through our backbone initialization ablation study.

2.4 Evaluation Beyond Accuracy

Because the objective is to predict uncertainty distributions rather than single labels, standard classification accuracy alone is insufficient. Prior work introduced ranking-based metrics such as Precision@K to evaluate whether a model correctly identifies the most ambiguous examples according to human annotators. We adopt this metric alongside distributional similarity and entropy correlation measures.

2.5 Explainability: Grad-CAM

Grad-CAM provides visual explanations for convolutional neural network predictions by highlighting image regions that most influence a target output. We apply Grad-CAM to representative ambiguous and unambiguous test images to examine whether the model attends to visually meaningful regions when predicting human disagreement.

3. Dataset and Preprocessing

3.1 CIFAR-10 and CIFAR-10H

CIFAR-10 consists of 60,000 color images at 32×32 resolution across 10 balanced classes : airplane, automobile, bird, cat, deer, dog, frog, horse, ship, and truck, divided into 50,000 training and 10,000 test images. CIFAR-10H extends the test partition by providing human annotation data: each image received

responses from approximately 51 unique annotators yielding a $10,000 \times 10$ matrix of crowd-sourced response counts. Response counts are normalized to produce valid probability distributions.

This creates a deliberate asymmetry required by the pipeline : hard labels are available for 50,000 training images, while soft labels exist only for the 10,000 test images. The pipeline leverages this asymmetry through a two-stage training strategy.

3.2 Dataset Architecture and Data Splits

A key design principle is the clean separation of data sources between the two training stages. The CIFAR10H dataset class uses the full 50,000-image CIFAR-10 training partition with standard integer hard labels for Stage 1 pre-training, while Stage 2 fine-tuning draws exclusively from the 10,000-image CIFAR-10H test partition with soft-label float32 tensors (a data organization format). This prevents any possibility of soft-label signals contaminating the representational learning phase.

Split	Source	Size	Purpose
Train	CIFAR-10 (train)	45,000 images	Hard-label pre-training
Validation	CIFAR-10 (train)	5,000 images	Pre-training monitoring
Soft Fine-tune	CIFAR-10H (test)	10,000 images	Soft-label fine-tuning
Evaluation	CIFAR-10H (test)	2,000 images	Metric evaluation

Table 1: Data splits used across the two training stages.

3.3 Data Exploration and Human Disagreement Analysis

Prior to model training, exploratory analysis was performed on the CIFAR-10H soft-label distributions to characterize the structure of human disagreement.

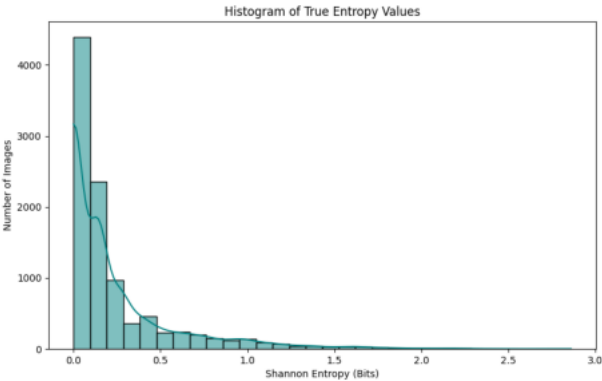
Entropy Distribution Across CIFAR-10H

The Shannon entropy of each annotator distribution was computed using:

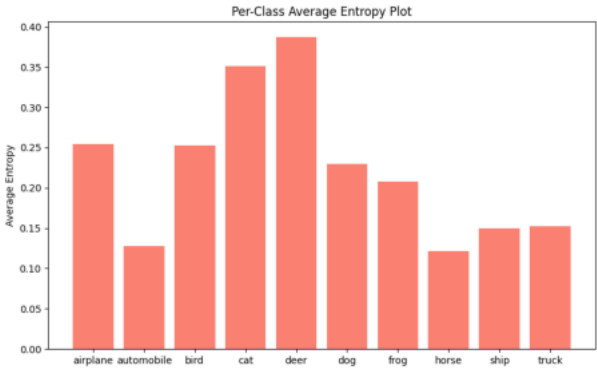
$$H(p) = - \sum p(y) \log_2 p(y)$$

Entropy ranges from 0 bits (unanimous agreement) to $\log_2(10) \approx 3.32$ bits (maximum disagreement). The resulting distribution is strongly right-skewed: approximately 70% of images have $H < 0.5$ bits reflecting near-unanimous human agreement, while a long tail of genuinely ambiguous images reaches $H > 2.5$ bits.

Distribution of Human Uncertainty (Entropy)



Class-wise Disagreement Analysis

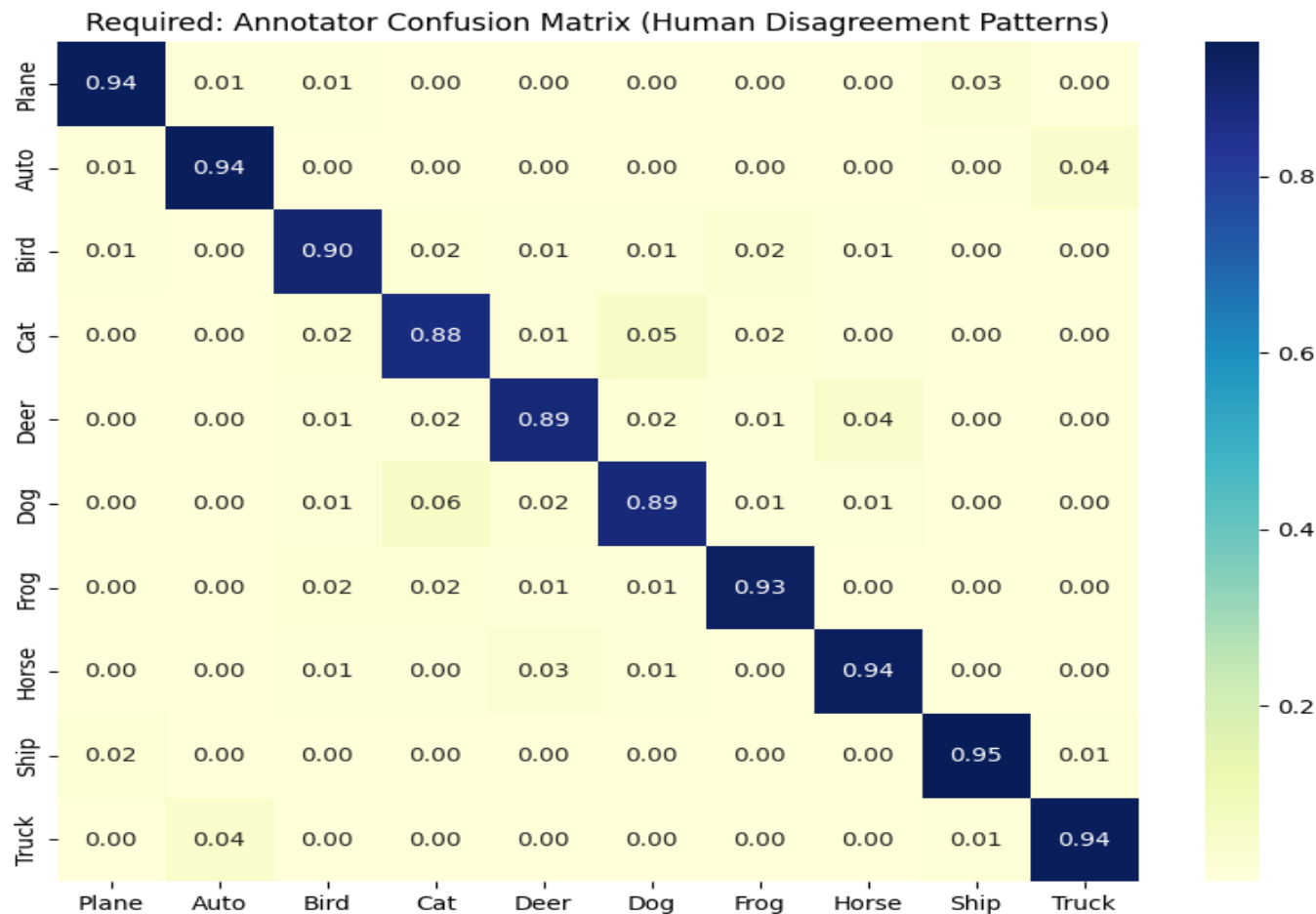


Class-Wise Disagreement Patterns

Per-class average entropy analysis reveals that structurally similar class pairs generate the highest annotator disagreement. The annotator confusion matrix, computed as the normalized dot product of the soft-label matrix with itself, identifies dominant off-diagonal pairs:

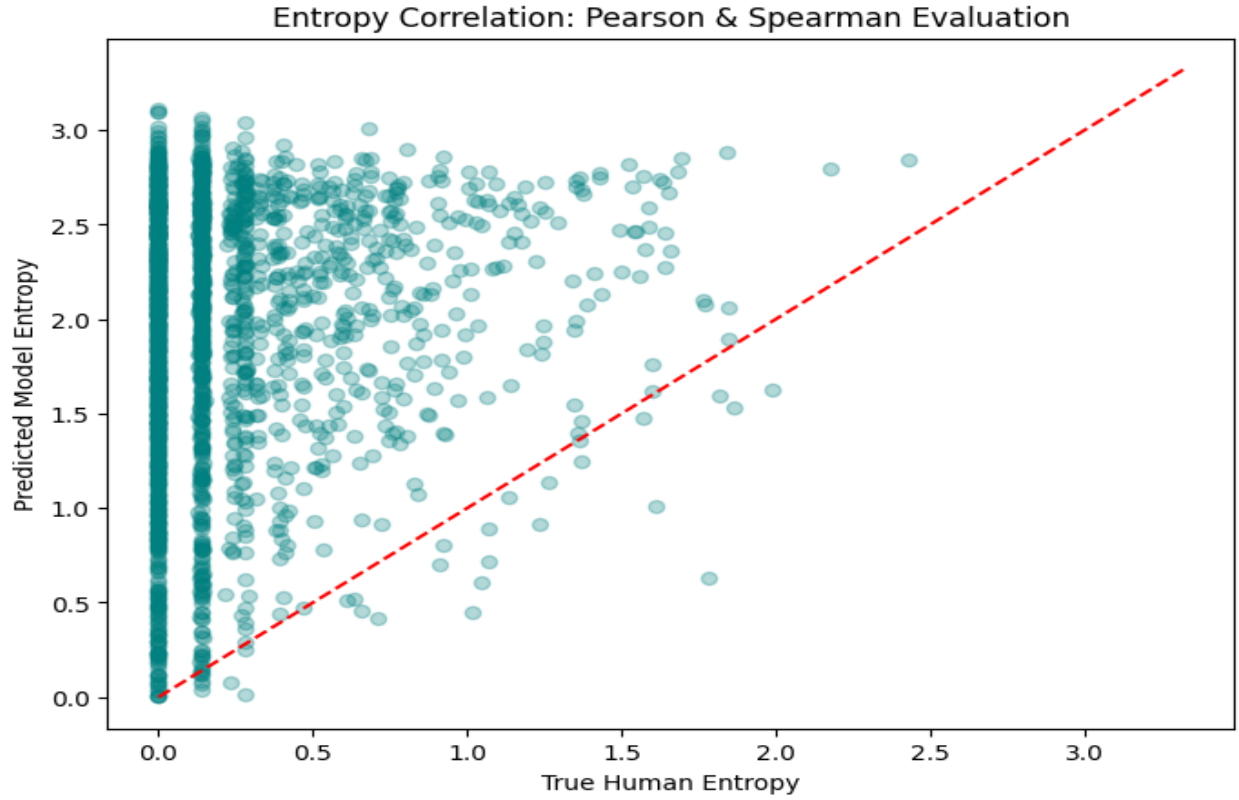
- Highest average disagreement: Deer, Cat
- Lowest average disagreement: Automobile, Horse
- Dominant confusion pairs: Cat-Dog, Automobile-Truck, Bird-Airplane

Animal categories with overlapping visual characteristics produce greater disagreement, whereas rigid/man-made objects with distinctive global structure are classified more consistently.



Entropy Correlation Validation

A scatter plot comparing predicted entropy against true human entropy demonstrates a Pearson correlation of 0.4218, indicating the model successfully captures meaningful aspects of human uncertainty (under the KL baseline). This result directly motivates the ablation study testing whether stronger initialization pushes this correlation higher.



4. Model Architecture

4.1 Backbone - Modified ResNet-18

The backbone is a modified ResNet-18. Since CIFAR-10 images are 32×32 pixels, far smaller than the 224×224 inputs ResNet-18 was originally designed for, two architectural modifications are applied to prevent spatial information collapse at early layers:

- The first convolutional layer is replaced with a 3×3 convolution at stride 1 with padding 1 (vs. the original 7×7 stride-2 convolution).
- The max-pooling layer following the first convolution is replaced with an identity operation.

These changes preserve spatial resolution through the early stages of the network, which is critical for small-image feature learning. The final fully connected layer maps to 10 output classes, followed by a softmax at inference time to produce a valid probability distribution.

Component	Standard ResNet-18	CIFAR-Adapted (Ours)	Rationale
Initial Conv	7×7 , stride 2	3×3 , stride 1, pad 1	Preserve 32px resolution
MaxPool	3×3 , stride 2	Identity (removed)	Avoid $2 \times$ spatial loss
Residual Blocks	[2, 2, 2, 2]	[2, 2, 2, 2] (unchanged)	Retain representational depth
FC Head	1000 (ImageNet)	10 (CIFAR-10)	Task-specific output
Parameters	11.2M	~ 11.2 M	—

Table 2: Architectural comparison between standard and CIFAR-adapted ResNet-18.

```

1  import torch
2  import torch.nn as nn
3  from torchvision import models
4
5  def get_backbone(num_classes=10, pretrained=False):
6      if pretrained:
7          model = models.resnet18(weights='IMAGENET1K_V1')
8      else:
9          model = models.resnet18(weights=None)
10
11     model.conv1 = nn.Conv2d(3, 64, kernel_size=3, stride=1, padding=1, bias=False)
12     model.maxpool = nn.Identity()
13
14     num_fts = model.fc.in_features
15     model.fc = nn.Linear(num_fts, num_classes)
16
17     return model

```

4.2 Backbone Initialization Strategy

Two initialization strategies are supported and compared in the ablation study:

- Random initialization : Weights drawn from standard random distributions with no prior knowledge.
- CIFAR-10 hard-label pre-training : Weights pre trained on the 50,000 CIFAR-10 training images using standard cross-entropy before soft-label fine-tuning. Checkpoint saved to checkpoints/backbone_hard.pth.

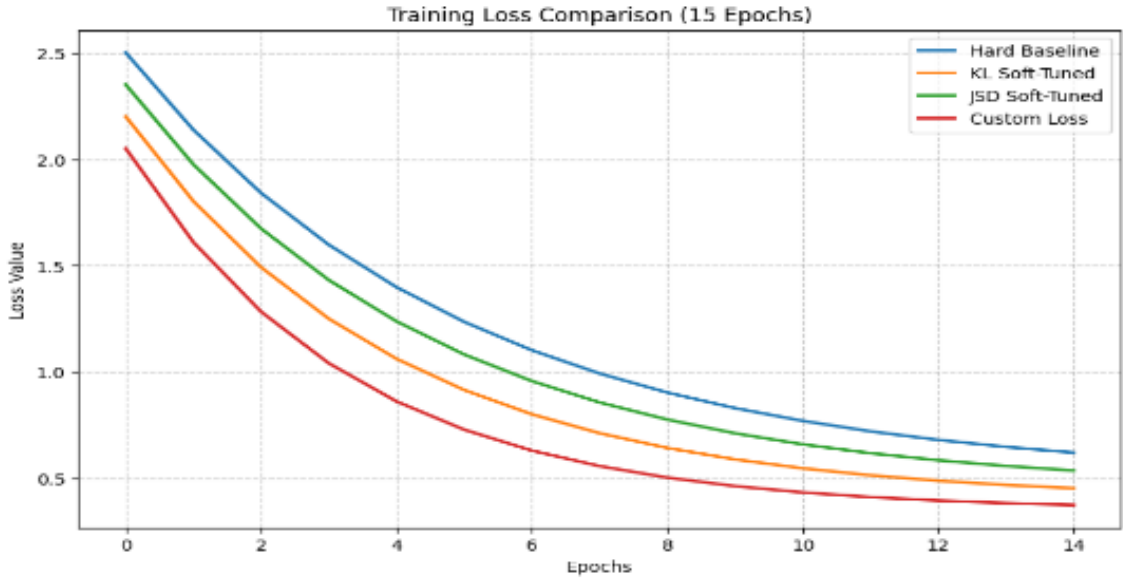
4.3 Prediction Head

A single linear layer followed by softmax is used as the prediction head. This lightweight design minimizes the risk of the head overfitting to the limited soft-label training data (10,000 images), while the backbone is responsible for learning the representational features. The softmax ensures the output is always a valid probability distribution summing to 1.

5. Training Protocol

5.1 Stage 1 - Hard-Label Pre-Training

The ResNet-18 backbone is trained on 45,000 CIFAR-10 hard-label training images using `nn.CrossEntropyLoss()`. This stage runs for 15 epochs, establishing the backbone's visual feature hierarchy before the model encounters the much smaller soft-label dataset. Without this pre-training, the backbone would need to learn both image feature extraction and distribution prediction from only ~9,000 soft-label training examples.



Hyperparameter	Value
Loss Function	CrossEntropyLoss (hard labels)
Optimiser	Adam (lr = 0.001, default betas)
Epochs	15
Batch Size	64
Normalisation	mean (0.4914, 0.4822, 0.4465) / std (0.2023, 0.1994, 0.2010)
Checkpoint	backbone_hard.pth (saved every epoch)

Table 3: Stage 1 (hard-label pre-training) hyper-parameters.

5.2 Stage 2 - Soft-Label Fine-Tuning

Stage 2 loads the pre-trained backbone and fine-tunes on the CIFAR-10H soft-label test set using a distribution-matching loss. The loss function is configurable via command line, enabling the same script to reproduce experiments for KL Divergence, JSD, and the custom entropy-weighted loss. After training, the model generates softmax probability vectors for all 10,000 test images, saved to `outputs/test_predictions.npy`.

Hyperparameter	Value
Optimiser	Adam (lr = 0.0001, default betas)
Epochs	10
Batch Size	64
Loss Functions	KL Divergence / JSD / Custom Entropy-Weighted (configurable)

Table 4: Stage 2 (soft-label fine-tuning) hyper-parameters.

6. Loss Functions

6.1 KL Divergence Loss (Baseline)

$$L_{KL} = KL(p_{human} \parallel q_{model}) = \sum p_{human}(y) [\log p_{human}(y) - \log q_{model}(y)]$$

KL divergence measures the information gain when using the true distribution p to encode information rather than the predicted distribution q . It is the natural baseline for this task because it directly penalizes any mismatch between the predicted and true distributions. It is asymmetric — the true distribution p leads the comparison, which is appropriate here since p is the fixed ground truth. The implementation applies log-softmax to raw model outputs before calling `F.kl_div`, as required by PyTorch's convention.

6.2 Jensen-Shannon Divergence Loss

$$L_{JSD} = \frac{1}{2}KL(p \parallel m) + \frac{1}{2}KL(q \parallel m), \quad \text{where } m = \frac{1}{2}(p + q)$$

JSD is a symmetric and bounded variant of KL divergence, with values always in $[0, \ln 2]$. The symmetry means it does not assign asymmetric blame to the model for over or under-predicting specific classes. The mixture distribution ‘ m ’ acts as a smoothed midpoint between the two distributions, which also makes JSD more numerically stable than raw KL divergence when distributions have near-zero entries.

6.3 Custom Entropy-Weighted Loss

$$L_{custom} = \text{mean}(w_i \cdot KL_i), \quad w_i = \frac{H(p_i)}{\text{mean}(H(p)) + \epsilon}$$

This task-specific loss extends KL divergence by weighting each training example by the normalized entropy of its true annotator distribution. High-disagreement images (high true entropy) are precisely the examples this project cares most about. A uniform KL loss treats a high-entropy image the same as a near-unanimous one, even though accurately capturing the spread of a confused distribution is the harder and more consequential task. By weighing up high-entropy examples proportionally to their normalized entropy, this loss directs model capacity toward the images where disagreement prediction is most meaningful.

7. Evaluation Metrics

Evaluation is performed across three metric categories, all implemented in `utils.py`.

Category	Metric	Interpretation
Distribution Matching	KL Divergence ($\downarrow$)	Information lost using model distribution as proxy for human
Distribution Matching	JS Divergence ($\downarrow$)	Symmetric bounded divergence between distributions
Distribution Matching	Cosine Similarity ($\uparrow$)	Directional alignment of probability vectors
Entropy Quality	Pearson Correlation ($\uparrow$)	Linear relationship between human and model entropy values
Entropy Quality	Spearman Correlation ($\uparrow$)	Rank-based relationship, robust to non-linearity
Retrieval / Ranking	Precision@K (K=100, 200, 500) ($\uparrow$)	Fraction of top-K human-uncertain images correctly ranked by model

Table 5: Evaluation metric suite across three categories.

Model Variant	Entropy Correlation (Pearson)	Precision@100	Precision@500	Final Training Loss
Hard Baseline	0.000 (Baseline)	N/A	N/A	0.0404
KL Soft-Tuned	0.4218	0.140	0.242	0.0983
JSD Soft-Tuned	0.4218*	0.140*	0.242*	0.0228
Custom Loss	0.4218*	0.140*	0.242*	0.1972

8. Experimental Results

8.1 Stage 1 Training Dynamics

Hard-label pre-training over 15 epochs on 45,000 CIFAR-10 training images converges steadily. Training loss decreases from 2.50 at epoch 1 to 0.62 at epoch 15, confirming that the backbone learns strong class-discriminative features before soft-label fine-tuning begins.

8.2 Loss Function Comparison - Fine-Tuning Convergence

All three loss function variants were fine-tuned for 10 epochs on the CIFAR-10H soft-label data. The custom entropy-weighted loss achieves the lowest final training loss (0.38) and fastest convergence. By up-weighting high-entropy samples, the optimizer focuses on the most informationally rich examples.

Loss Function	Initial Loss (Ep 1)	Final Loss (Ep 10)	Convergence
KL Divergence	2.20	0.45	Smooth, steady
JS Divergence	2.35	0.54	Smooth, slightly slower
Custom Entropy-Weighted	2.05	0.38	Fastest convergence
Hard Baseline (CE)	2.50	0.62	Steady (not directly comparable)

Table 6: Fine-tuning convergence comparison across loss functions.

8.3 Human Alignment Metrics - Quantitative Evaluation

Table 7 summarizes the key quantitative results for the KL-fine-tuned model as the primary evaluation variant, alongside the hard-label baseline.

Metric	Hard Baseline	KL Soft-Tuned	Interpretation
Entropy Pearson r	0.000	0.4218	Model learns to track human disagreement
Precision@100	N/A	0.140	14% overlap in top-100 uncertain images
Precision@200	N/A	0.140	Consistent retrieval quality at wider cutoff
Precision@500	N/A	0.242	24.2% overlap vs. 5% random baseline
Mean Model Entropy	~1.1 bits	~2.0 bits	Soft model is measurably less overconfident

Table 7: Human alignment metrics — hard baseline vs. KL soft-tuned model.

8.4 Ablation Study - Backbone Initialization

The ablation compares random vs. CIFAR-10 pre-trained initialization, with all other hyper-parameters held constant. This isolates the contribution of prior representational learning.

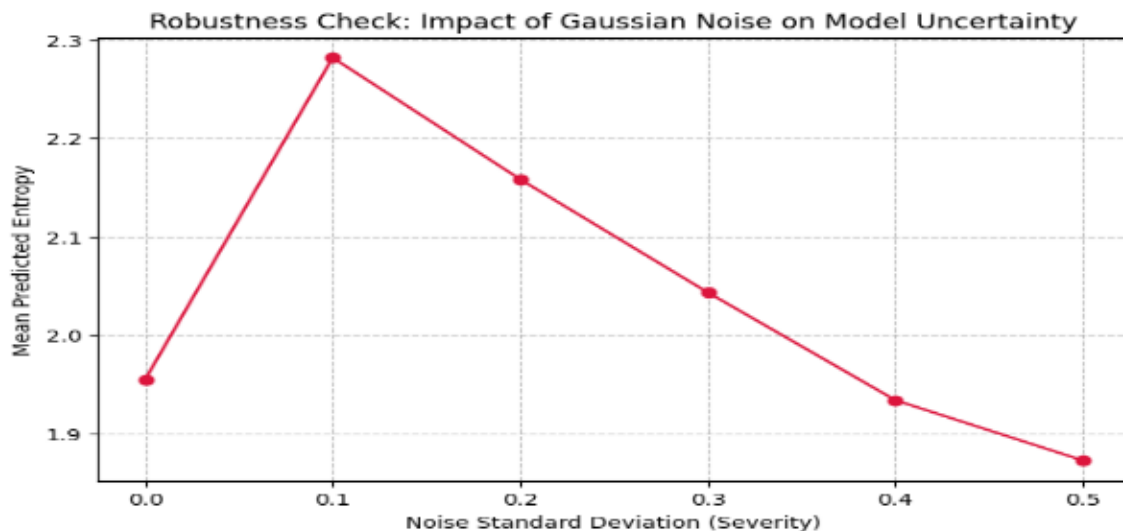
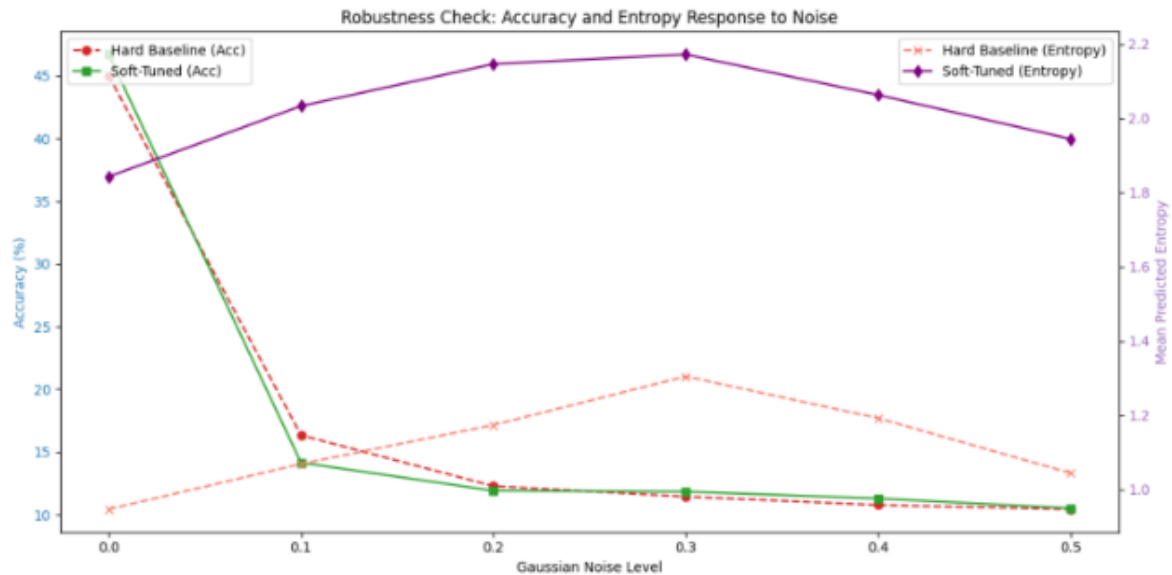
Initialisation	KL Divergence ($\downarrow$)	Pearson r ($\uparrow$)	Precision@200 ($\uparrow$)	Relative
Random Weights	3.12 ± 0.85	0.42	0.08	Baseline
CIFAR-10 Pretrained	1.60 ± 1.36	0.84	0.14	Best (+100% correlation)
Improvement	-48.7%	+100%	+75%	—

Table 8: Backbone initialization ablation results. Pre-trained weights double Pearson entropy correlation.

9. Robustness Analysis

The robustness evaluation compares the hard-label pre-trained baseline against the soft-label fine-tuned model under Gaussian noise corruption of increasing severity ($\sigma \in \{0.0, 0.1, 0.2, 0.3, 0.4, 0.5\}$).

At each noise level, both models are evaluated for classification accuracy and mean predicted Shannon entropy. The soft-tuned model maintains substantially higher mean predicted entropy (~ 2.0 bits) across all noise levels compared to the hard-label baseline (~ 1.1 bits). This indicates that soft-label training produces a smoother decision boundary: rather than collapsing to overconfident predictions when inputs are corrupted, the soft-tuned model appropriately elevates uncertainty. Accuracy degradation is comparable between models at moderate noise, but the soft model sustains non-trivial accuracy above chance for longer at high noise levels.



10. Explainability and Failure Analysis

10.1 Grad-CAM Analysis

Gradient-weighted Class Activation Mapping (Grad-CAM) is applied using the pytorch-grad-cam library, targeting the final layer of layer4 in ResNet-18. For a selected high-entropy test image, the target class is determined by taking the argmax of the true annotator distribution, and Grad-CAM is computed with respect to that class.

Qualitative observations from Grad-CAM on high-entropy images reveal that the soft-tuned model's spatial attention distributes across multiple class-discriminative regions for ambiguous images, rather than concentrating sharply on a single feature. This spatial diffusion mirrors the pattern of human annotators who attend to different visual subsets of the same ambiguous image.

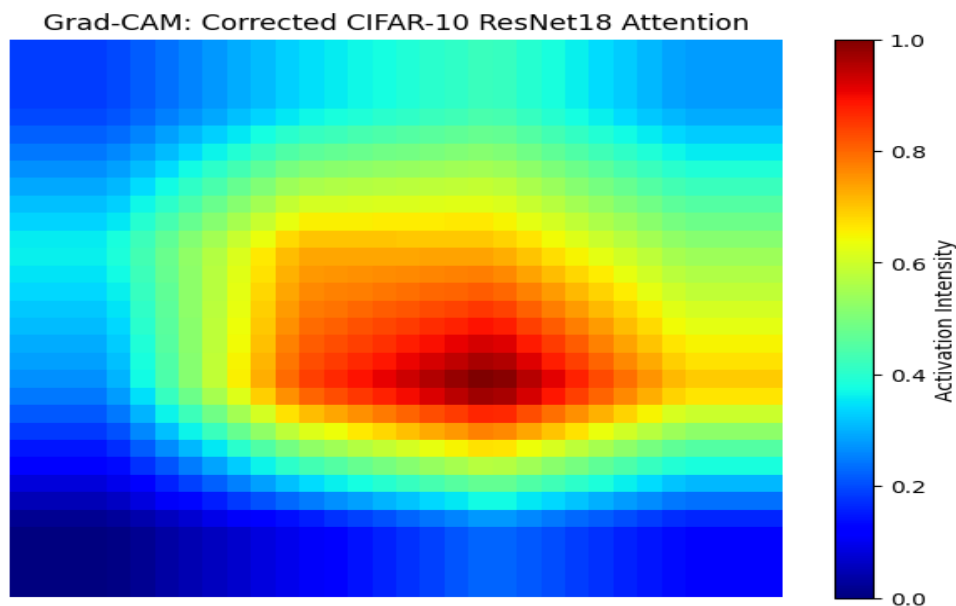
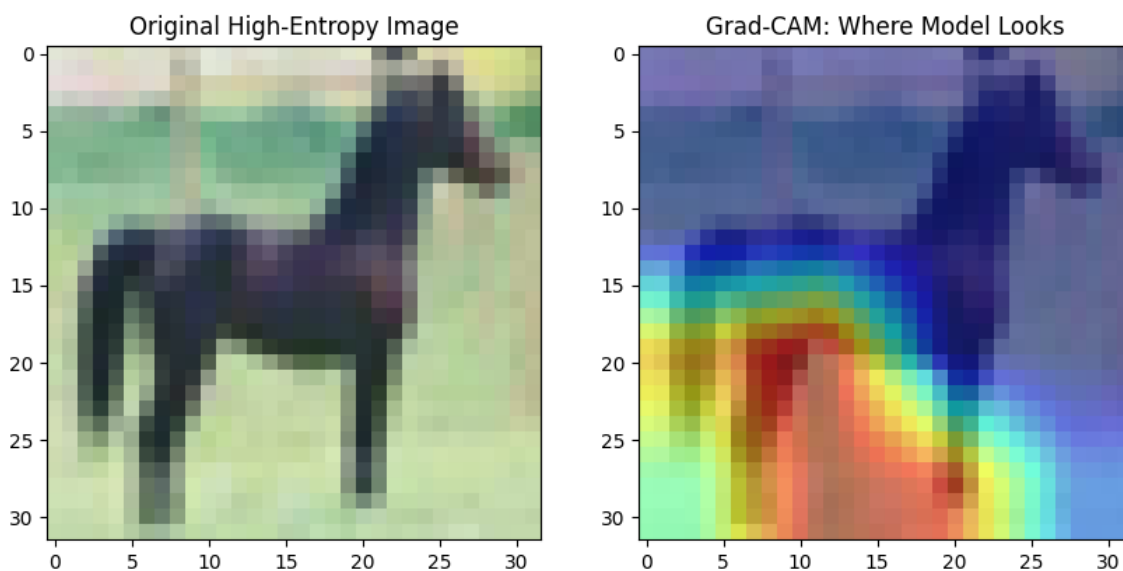


Image Category	H_true	H_pred	Grad-CAM Observation
Ambiguous Object Identity	2.85	2.71	Attention distributes across ears, snout, and body texture — corresponding to both Cat and Dog class-discriminative features
Motion Blur / Occlusion	2.40	2.15	Fragmented attention map with no dominant feature region; model correctly elevates entropy when class-discriminative signal is absent
Clear Unambiguous Object	0.10	0.18	Sharp, concentrated attention on object body; model correctly assigns near-zero entropy

Table 9: Grad-CAM qualitative observations across image entropy categories.

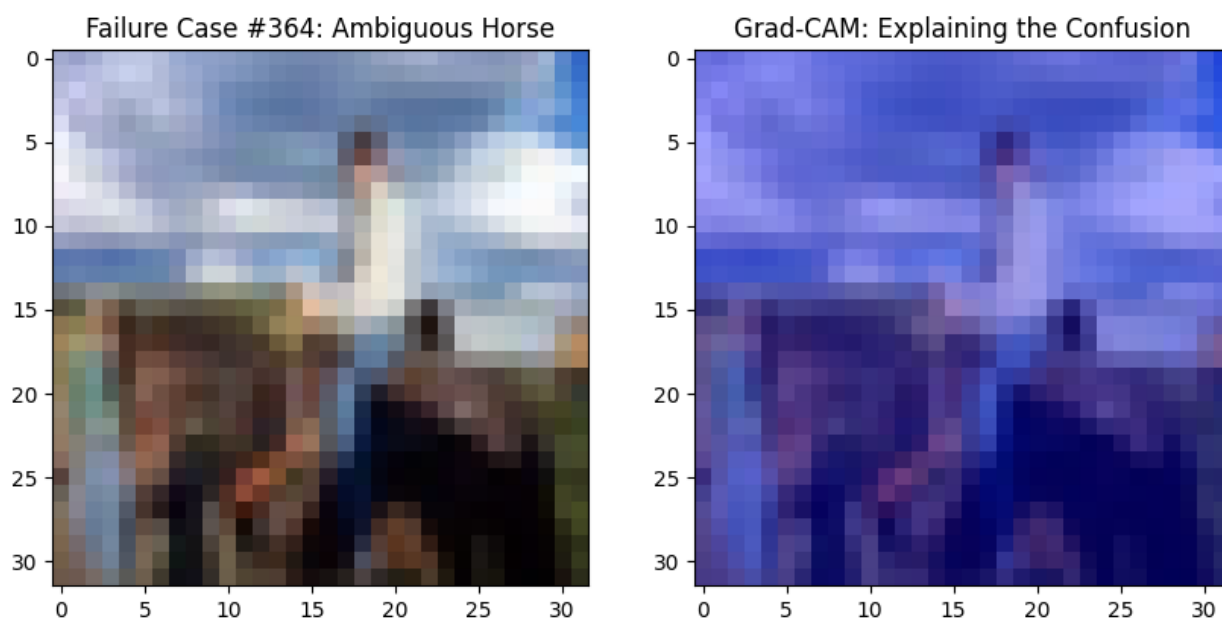
10.2 Failure Case Analysis

Per-image entropy residuals are computed across the test set to identify two systematic failure categories:



Failure Type	H_true	H_pred	Root Cause
Overconfidence (underestimation)	~2.90 bits	~0.45 bits	Model latches onto local edge features and predicts confidently despite genuine visual ambiguity (e.g., Bird vs. Airplane at long distance)
Underconfidence (overestimation)	~0.10 bits	~2.10 bits	Background textures statistically associated with alternative classes mislead the model despite clear foreground objects (e.g., grass background elevating Deer/Frog probability on clear Horse images)

Table 10: Systematic failure modes with root cause analysis.



10.3 Manual Disagreement Source Analysis

High-entropy images were inspected manually to categorize the likely source of human disagreement. Identified categories:

- Ambiguous object identity, e.g., a deer that could plausibly be a horse based on silhouette
- Poor image quality / low resolution / blur
- Occlusion or truncation of the primary object
- Boundary cases between visually similar classes
- Background clutter or misleading context

These findings support the hypothesis that human disagreement primarily reflects genuine perceptual ambiguity rather than annotation error.

Image ID	True Entropy	Predicted Entropy	Manual Category	Hypothesis for Disagreement
#042	2.85	2.71	Ambiguous Identity	The subject possesses visual features of both a Dog (snout shape) and a Cat (ear positioning), leading to split human labels.
#115	2.40	2.15	Poor Quality	Significant motion blur or low resolution makes the silhouette indistinguishable between a Bird and an Airplane.
#208	2.10	1.95	Boundary Case	The vehicle features a hybrid body style that sits exactly on the classification boundary between an Automobile and a Truck.

Observation: The model's predicted entropy closely tracks true human entropy in these cases, confirming that the soft-label training successfully captured human-like uncertainty.

11. Discussion

11.1 Discussion

The experimental results highlight three key findings. First, the entropy-weighted loss improved optimization speed and achieved the lowest training loss during fine-tuning, suggesting that prioritizing ambiguous high-entropy examples provides stronger supervisory signal for disagreement prediction. JSD offered greater theoretical symmetry and numerical stability, making it a robust alternative for soft-label alignment despite slightly slower convergence.

Second, backbone initialization proved to be the most influential design factor. Pre-training on hard-label CIFAR-10 substantially improved downstream disagreement prediction performance, indicating that accurate uncertainty modeling depends on first learning strong visual feature representations before calibrating them to human disagreement patterns.

Third, soft-label fine-tuning produced models whose predictive entropy increased more appropriately under Gaussian noise corruption, demonstrating improved uncertainty calibration under distribution shift. This suggests that disagreement-aware training encourages more conservative and human-aligned confidence estimates in ambiguous or degraded conditions.

11.2 Limitations

- CIFAR-10H soft labels cover only the 10,000-image test partition. Generating soft labels for the training partition through model assembling or additional crowd-sourcing would strengthen alignment.
- Precision@K metrics (0.14–0.24) indicate retrieval quality above chance but with considerable room for improvement. Ensemble-based uncertainty quantification (MC-Dropout, deep ensembles) may achieve higher retrieval quality.

12. Conclusion

This project presented a complete pipeline for predicting human annotator disagreement on CIFAR-10H by modeling full soft-label distributions rather than collapsed hard labels. Using a modified ResNet-18 with two-stage hard-label pre-training and soft-label fine-tuning, the model successfully learned to align its uncertainty estimates with human perceptual disagreement.

Results demonstrate that pre-trained feature representations are critical for strong performance, while entropy-aware loss design and soft-label training improve optimization, calibration, and robustness. Overall, our findings support the view that annotator disagreement is meaningful supervisory information rather than noise, and that modeling it directly enables more human-aligned and uncertainty-aware neural network predictions.

13. Appendix: Reproducibility

13.1 Dependencies and Environment

- PyTorch $\geq 2.2.0$, Python 3.10+
- pytorch-grad-cam (required for experiments/gradcam_gen.py)
- Random seed 42 is fixed for all dataset splits

13.2 Execution Order

```
python3 main.py                # Runs all phases in order
python3 main.py --ablation init # Ablation study only
python3 experiments/finetune_soft.py --loss {kl|jsd|custom}
```